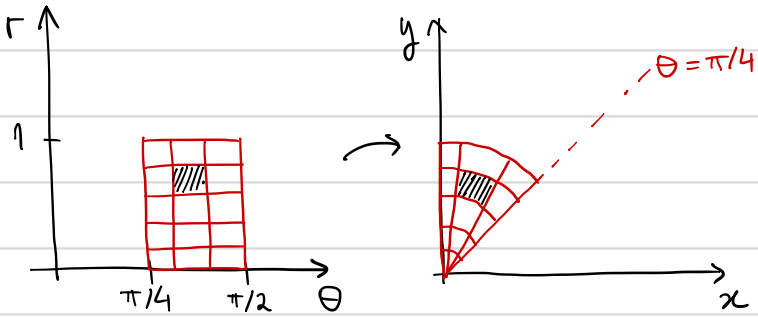
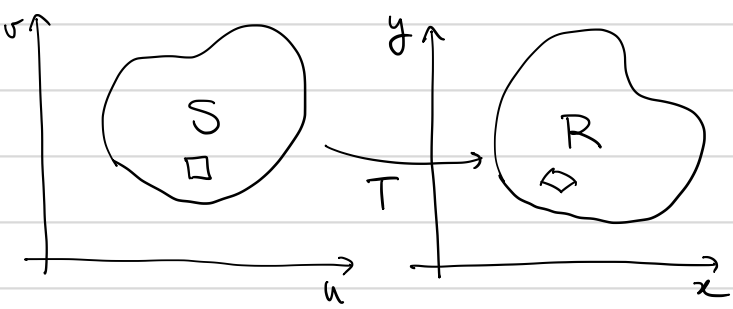
Change of variables (15.9)

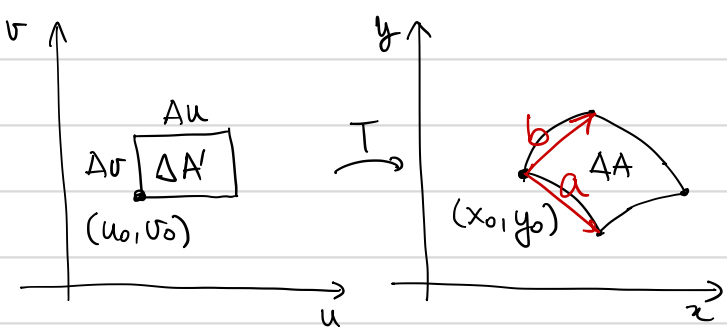


$$dx\,dy = r\,dr\,d\theta$$

In general: Find scaling factor
($dx\,dy$ vs $du\,dv$)



Zoom:



$$T(u,v) = (x,y)$$

$$\vec{a} = T(u_0+\Delta u, v_0) - T(u_0, v_0)$$
$$\vec{b} = T(u_0, v_0+\Delta v) - T(u_0, v_0)$$

$$\vec{a} \approx \frac{\partial T}{\partial u}(u_0, v_0)\,\Delta u$$

$$= \frac{\partial x}{\partial u}\Delta u\,\hat{i} + \frac{\partial y}{\partial u}\Delta u\,\hat{j}$$

$$\vec{b} \approx \frac{\partial x}{\partial v}\Delta v\,\hat{i} + \frac{\partial y}{\partial v}\Delta v\,\hat{j}$$

$\pm$det = area of paralelogram

$$\Delta A \approx \pm \det\begin{bmatrix} \frac{\partial x}{\partial u}\Delta u & \frac{\partial y}{\partial u}\Delta u \\ \frac{\partial x}{\partial v}\Delta v & \frac{\partial y}{\partial v}\Delta v \end{bmatrix}$$

$$= \pm \underbrace{\frac{\partial(x,y)}{\partial(u,v)}}_{\frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial y}{\partial u}\frac{\partial x}{\partial v}}\Delta u\,\Delta v$$

"Jacobian determinant"

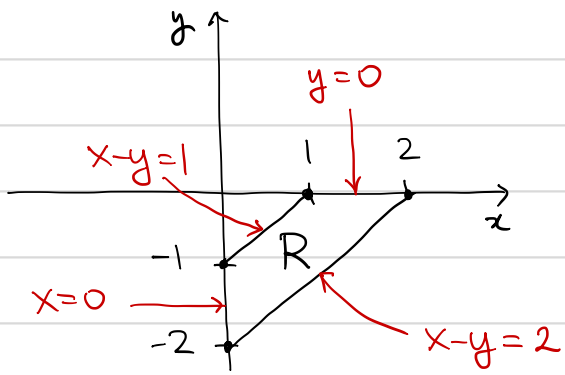
$$\boxed{dA = dx\,dy = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv}$$

Example: $x = r\cos\theta$
$y = r\sin\theta$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$= r(\cos^2\theta + \sin^2\theta) = \underline{\underline{r}}$$

Example: $\iint_R e^{(x+y)/(x-y)}\, dA$



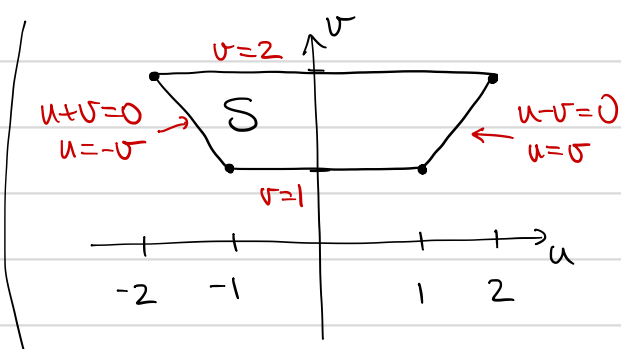
$u = x+y \qquad v = x-y$

$u+v = 2x \implies x = \frac{1}{2}u + \frac{1}{2}v$

$u-v = 2y \implies y = \frac{1}{2}u - \frac{1}{2}v$

$$\frac{\partial(x,y)}{\partial(u,v)} = \det\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} = -\frac{1}{2}$$

$$\iint_R e^{(x+y)/(x-y)}\, dA = \iint_S e^{u/v}\, \frac{1}{2}\, du\, dv$$



$$= \int_1^2 \int_{-v}^{v} e^{u/v}\, \frac{1}{2}\, du\, dv$$

$$\int_1^2 \int_{-v}^{v} e^{u/v}\, \frac{1}{2}\, du\, dv =$$

$$= \frac{1}{2}\int_1^2 \left[v e^{u/v} \right]_{u=-v}^{v} dv$$

$$= \frac{1}{2}\int_1^2 \left(v \cdot e^1 - v \cdot e^{-1} \right) dv$$

$$= \frac{1}{2}(e - e^{-1}) \left[\frac{v^2}{2} \right]_1^2 = \frac{3}{4}\left(e - \frac{1}{e}\right) //$$

Triple integrals: $x = x(u,v,w)$

$y = y(u,v,w)$

$z = z(u,v,w)$

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

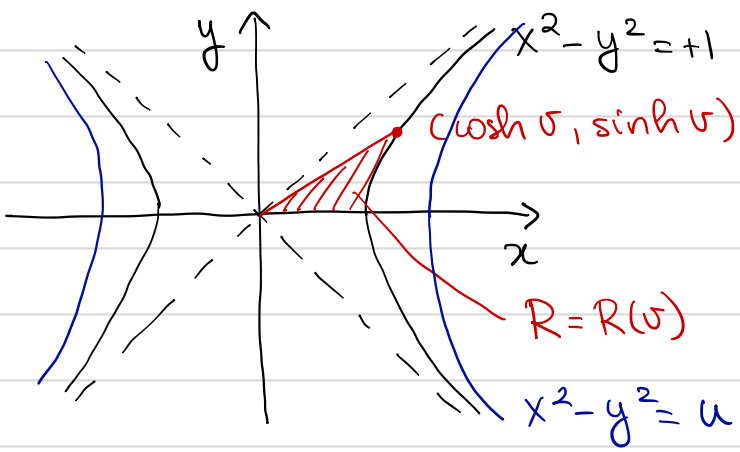
$$\boxed{dx\, dy\, dz = \left| \frac{\partial(x,y,z)}{\partial(u,v,w)} \right| du\, dv\, dw}$$

Supplementary notes

Hyperbolic coordinates

$$\begin{cases} x = \sqrt{u}\cosh v & u > 0 \\ y = \sqrt{u}\sinh v & v \in \mathbb{R} \end{cases}$$

$$\left[\begin{array}{ll} \cosh v = \dfrac{e^{v}+e^{-v}}{2} & \sinh v = \dfrac{e^{v}-e^{-v}}{2} \\ \underline{\text{Key}}: \cosh^2 v - \sinh^2 v = 1 & \end{array}\right]$$



$u =$ which hyperbola

$v =$ where in the hyperbola

$= 2 \cdot \text{Area}(R)$

$$\begin{bmatrix}\text{Area of} \\ R(v_0)\end{bmatrix} = \int_0^{v_0}\int_0^1 \underbrace{\frac{\partial(x,y)}{\partial(u,v)}}_{\frac{1}{2}}\, du\, dv = \frac{v_0}{2}$$

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{1}{2}u^{-1/2}\cosh v & u^{1/2}\sinh v \\ \frac{1}{2}u^{-1/2}\sinh v & u^{1/2}\cosh v \end{vmatrix}$$

$$= \frac{1}{2}\left(\cosh^2 v - \sinh^2 v\right) = \frac{1}{2} \,//$$